14. The sum which amounts to ₹2662 at 10% p.a in 3 years, compounded yearly is_____.

(A) ₹2000 (B) ₹1800 (C) ₹1500 (D) ₹2500

15. The difference between compound and simple interest on a certain sum of money for 2 years at 2% p.a is ₹1. The sum of money is _____ .

(A) ₹2000 (B) ₹1500 (C) ₹3000 (D) ₹2500

## 4.5 Compound Variation

Before we could learn about compound variation, let us recall the concepts on direct and inverse proportions.

If two quantities are such that an increase or decrease in one quantity makes a corresponding increase or decrease (same effect) in the other quantity, then they are said to be in direct proportion or said to vary directly. In other words, $x$ and $y$ are said to vary directly if $\boldsymbol{y=kx}$ always, where $k$ is called the proportionality constant and $k > 0$ assuming that $y$ depends on $x$ and so $\boldsymbol{k = \frac{y}{x}}$.

For example, let us assume that one of you plan to give 2 pens to each of your friends in the birthday party. Then the number of pens to be bought will be in direct proportion with the number of friends who will attend the party. Isn't it? The following table will help you understand this clearly.

| Number of friends ($x$) | 1 | 2 | 5 | 12 | 15 |
|---|---|---|---|---|---|
| Number of pens ($y$) | 2 | 4 | 10 | 24 | 30 |

In this case, we find that the proportionality constant, $k = \frac{y}{x} = \frac{2}{1} = \frac{4}{2} = \frac{10}{5} = \frac{24}{12} = \frac{30}{15} = 2$.

**Few more examples of Direct Proportion:**

1. Distance –Time (under constant speed): If the distance increases, then the time taken to reach that distance will also increase and vice- versa.
2. Purchase – Spending: If the purchase on utilities for a family during the festival time increases, then the spending limit also increases and vice versa.
3. WorkTime – Earnings: If the number of hours worked is less, then the pay earned will also be less and vice-versa.

Similarly, if two quantities are such that an increase or decrease in one quantity makes a corresponding decrease or increase (opposite effect) in the other quantity, then they are said to be in inverse (indirect) proportion or said to vary inversely. In other words, $x$ and $y$ are said to vary inversely, if $\boldsymbol{xy = k}$ always, where $k$ is called proportionality constant and $k > 0$.

For example, let us assume that a class of 30 students in a school walks on streets in a village for health awareness campaign in an orderly manner, then we can see an inverse proportion in the number of rows and columns they walk. The following table will help you understand this clearly.

| Number of students in columns ($x$) | 1 | 2 | 3 | 5 | 6 |
|---|---|---|---|---|---|
| Number of students in rows ($y$) | 30 | 15 | 10 | 6 | 5 |



04_8th_Maths_Life Mathematics_Chap 4.indd 140 20/11/2022 13:13:39